

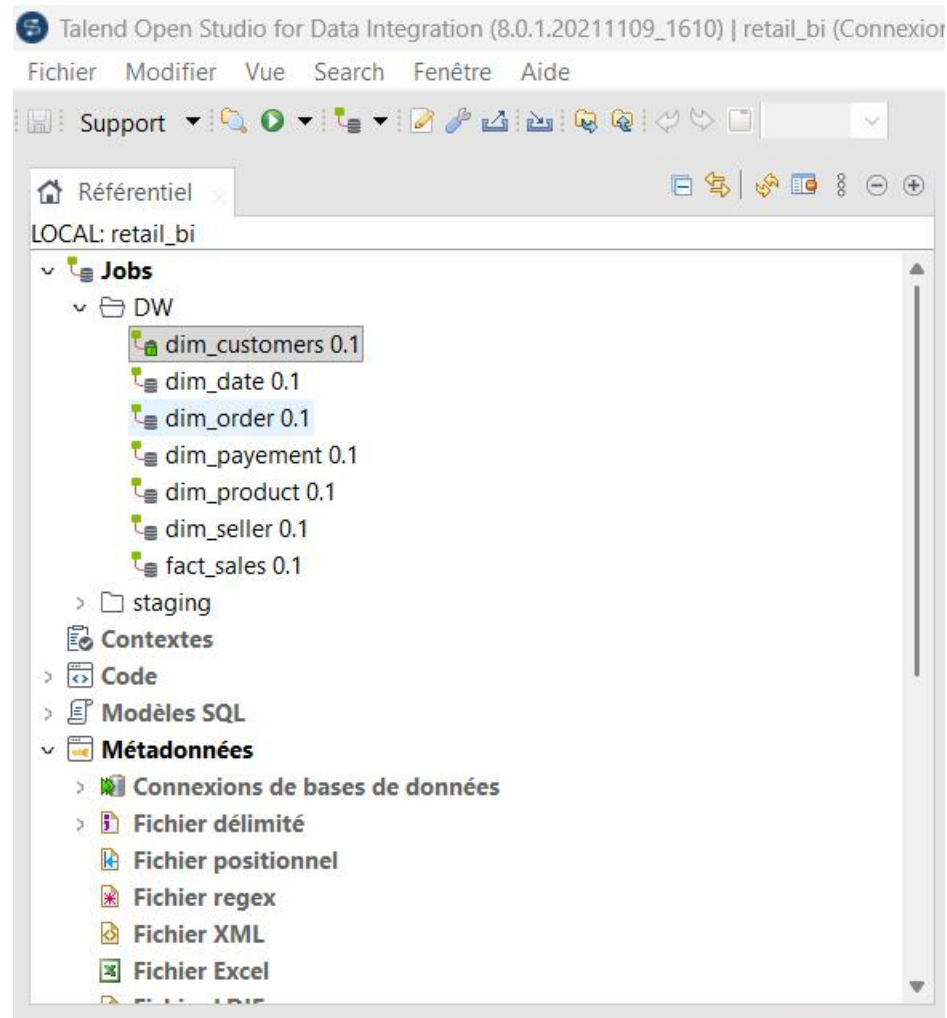
ETL Pipeline Documentation

Olist BI Retail Project

Data Extraction, Transformation & Loading using Talend and PostgreSQL



1. ETL Job Architecture



2.Customer dimention job ETL

Talend Open Studio for Data Integration (8.0.1.20211109_1610) | retail_bi (Connexion: Local)

LOCAL: retail_bi

Jobs

- dim_customers_0.1
- dim_date 0.1
- dim_order 0.1
- dim_payment 0.1
- dim_product 0.1
- dim_seller 0.1
- fact_sales 0.1

staging

Contextes

Code

Modèles SQL

Metadonnées

- Connexions de bases de données
- Fichiers délimités
- Fichiers positionnels
- Fichiers regex
- Fichiers XML
- Fichiers Excel

Outline

- tdBInput_1 ("customers_stg")
- tdBOutput_1 ("dim_customer")
- tMap_1
- tUniqueRow_1

Job dim_customers_0.1

Principal

Nom: dim_customers

Créé par: user@talend.com

Version: 0.1

Création: 12/08/2026 13:25

Modification: 20/08/2026 01:46

Objectif:

Description:

1 Item selected

Talend Open Studio for Data Integration - tMap - tMap_1

Find:

Var:

Auto map !

row2

Column

- customer_id
- customer_unique_id
- customer_zip_code_prefix
- customer_city
- customer_state

etl

Expression

Column

- row2.customer_id
- customer_id
- row2.customer_unique_id
- customer_unique_id
- row2.customer_zip_code_prefix
- customer_zip_code_prefix
- row2.customer_city
- customer_city
- row2.customer_state
- customer_state

row2

Colonne	Clé	Type	N.	Modèle de date (C...	Longueur	Précision	Par déf...	Comment...
customer_id	☐	String	☒		32	0		
customer_unique_id	☐	String	☒		32	0		
customer_zip_code_prefix	☐	String	☒		5	0		
customer_city	☐	String	☒		100	0		
customer_state	☐	String	☒		2	0		

etl

Colonne	Clé	Type	N.	Modèle de date (C...	Longueur	Précision	Par déf...	Comment...
customer_id	☐	String	☐		50	0		
customer_unique_id	☐	String	☒		50	0		
customer_zip_code_prefix	☐	Integer	☒		10	0		
customer_city	☐	String	☒		100	0		
customer_state	☐	String	☒		10	0		

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3.Date dimention etl job

The screenshot shows the Talend Open Studio interface for a job named 'dim_date 0.1'. The job is designed to create a date dimension table. The main workflow consists of the following components:

- tUniqRow_1**: Reads data from 'order_stg' and filters for 'row1 (Main)'.
- tMap_1**: A join component that combines 'row1 (Main)' with 'row2 (Uniques)'.
- etl (Main)**: An ETL component that processes the joined data.
- tUniqRow_2**: Writes the processed data to the 'dim_date' table, ensuring uniqueness.

The 'dim_date 0.1' component properties are shown below:

Principal	Nom	dim_date
Extra	Créé par :	user@talend.com
Stats & Logs	Création	12/08/2026 15:28
Version	Modification	20/08/2026 01:46
	Objectif	
	Description	

The screenshot shows the 'tMap_1' component editor in Talend Open Studio. It displays the mapping between the source data (row2) and the target data (etl) for the 'dim_date' table.

row2 Columns:

Column
order_id
customer_id
order_status
order_purchase_timestamp
order_approved_at
order_delivered_carrier_date
order_delivered_customer_date
order_estimated_delivery_date

etl Columns:

Expression	Column
TalendDate.parseDate(... "yyyy-MM-dd", ... Ta...	full_date
Integer.parseInt(TalendDate.formatDate("dd", row...	day
TalendDate.formatDate("EEEE", row2.order_purch...	day_name
Integer.parseInt(TalendDate.formatDate("MM", ro...	month
TalendDate.formatDate("MMMM", row2.order_pur...	month_name
((Integer.parseInt(TalendDate.formatDate("MM", r...	quarter
Integer.parseInt(TalendDate.formatDate("yyyy", ro...	year
Integer.parseInt(TalendDate.formatDate("ww", ro...	week_of_year

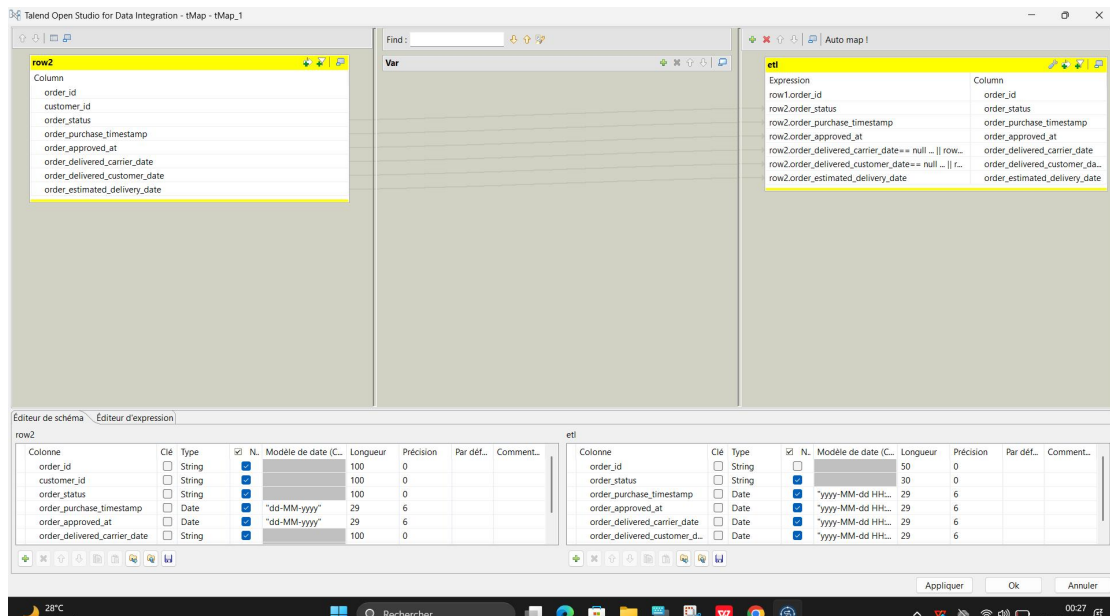
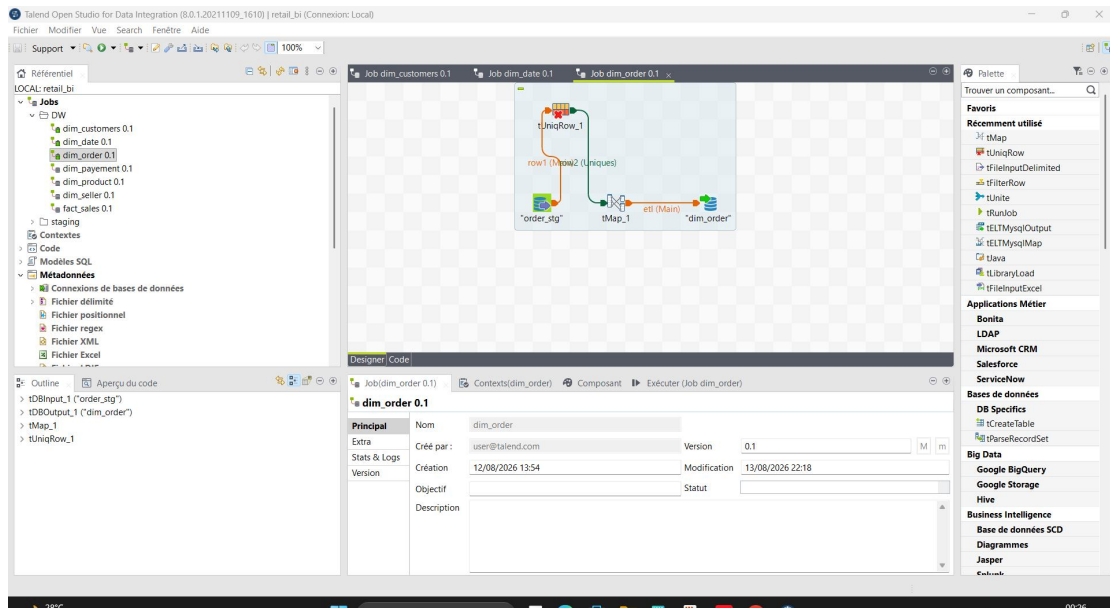
row2 Data Type Definitions:

Colonne	Clé	Type	N.	Modèle de date (C...	Longueur	Précision	Par déf...	Comment...
order_id		String	☒		100	0		
customer_id		String	☒		100	0		
order_status		String	☒		100	0		
order_purchase_timestamp		Date	☒	"dd-MM-yyyy"	29	6		
order_approved_at		Date	☒	"dd-MM-yyyy"	29	6		
order_delivered_carrier_date		String	☒		100	0		

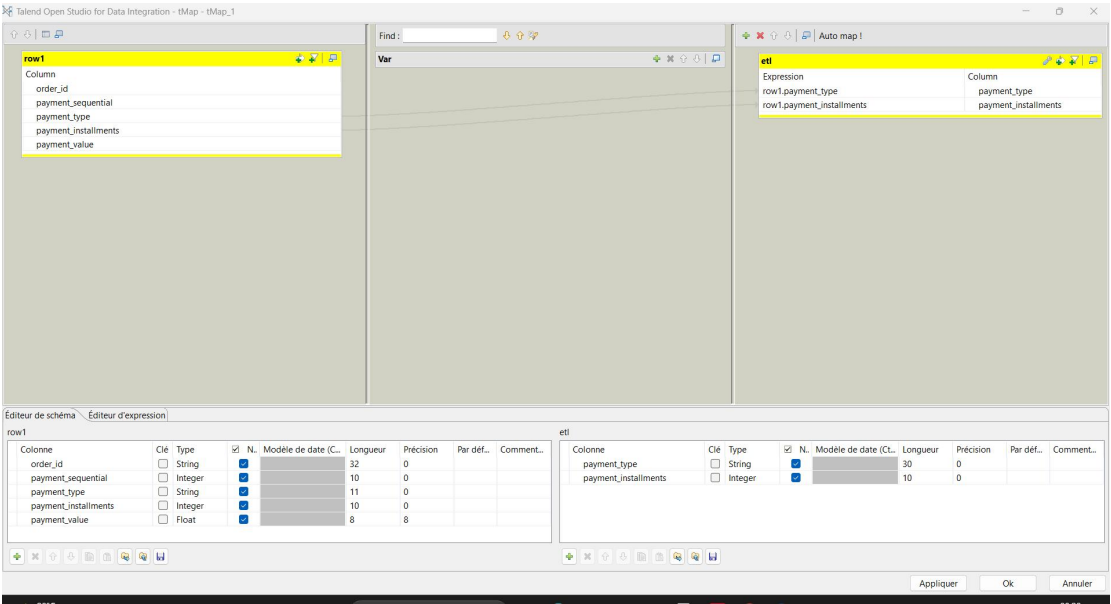
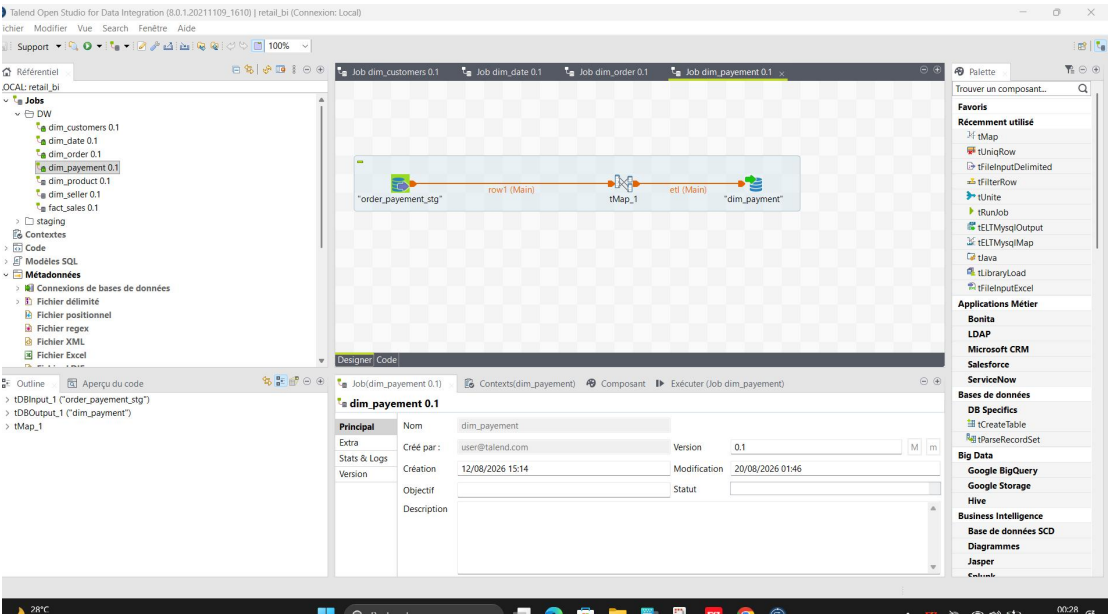
etl Data Type Definitions:

Colonne	Clé	Type	N.	Modèle de date (C...	Longueur	Précision	Par déf...	Comment...
full_date		Date	☐	"dd-MM-yyyy"	13	0		
day		Integer	☒		10	0		
day_name		String	☒		20	0		
month		Integer	☒		10	0		
month_name		String	☒		20	0		
quarter		Integer	☒		10	0		

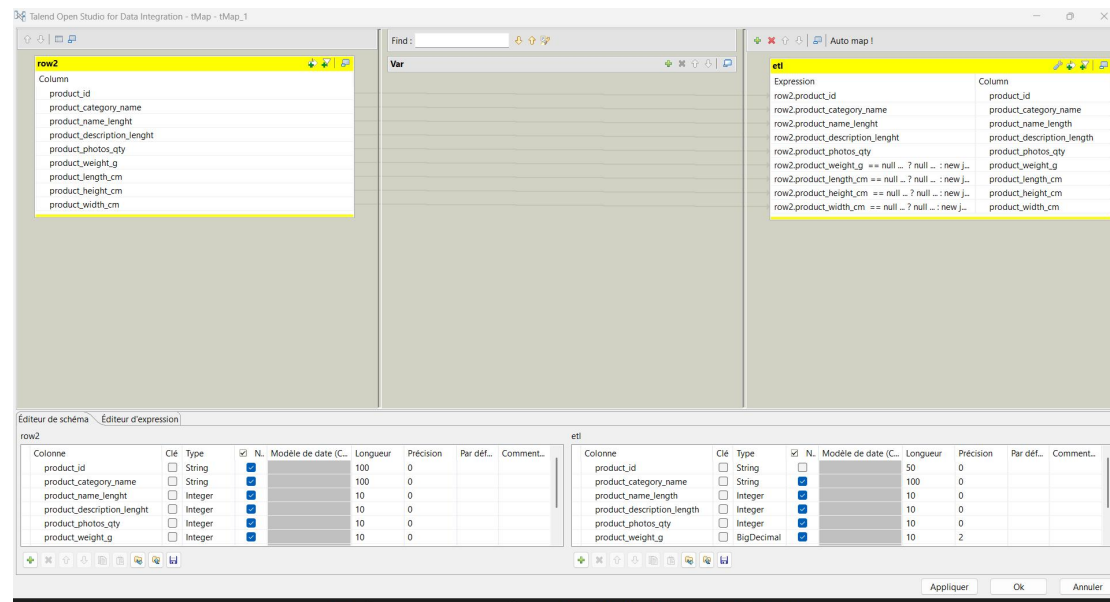
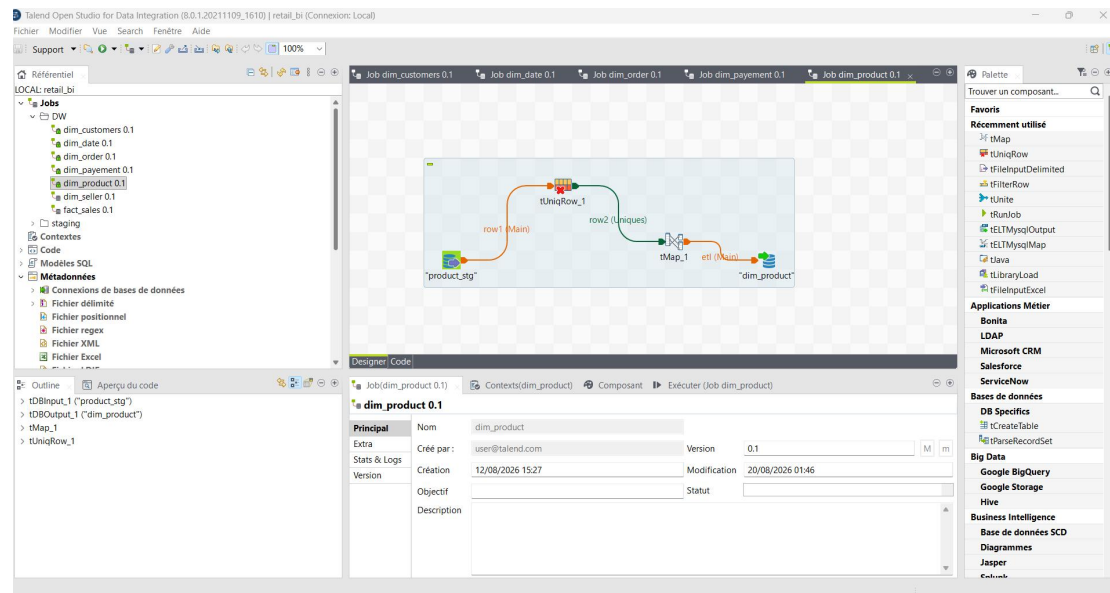
4. Order dimension etl job



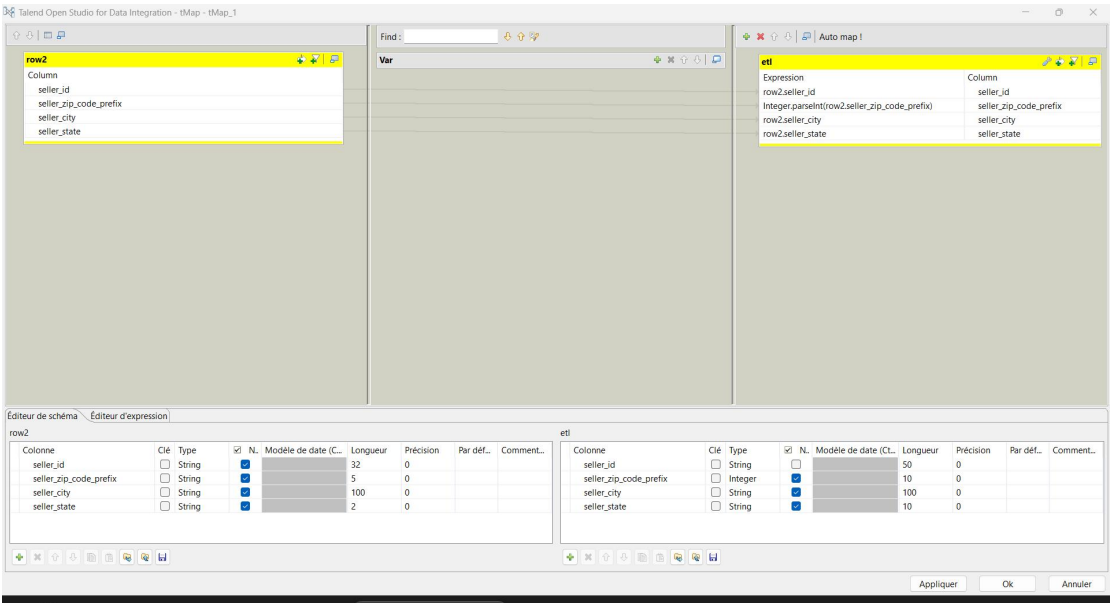
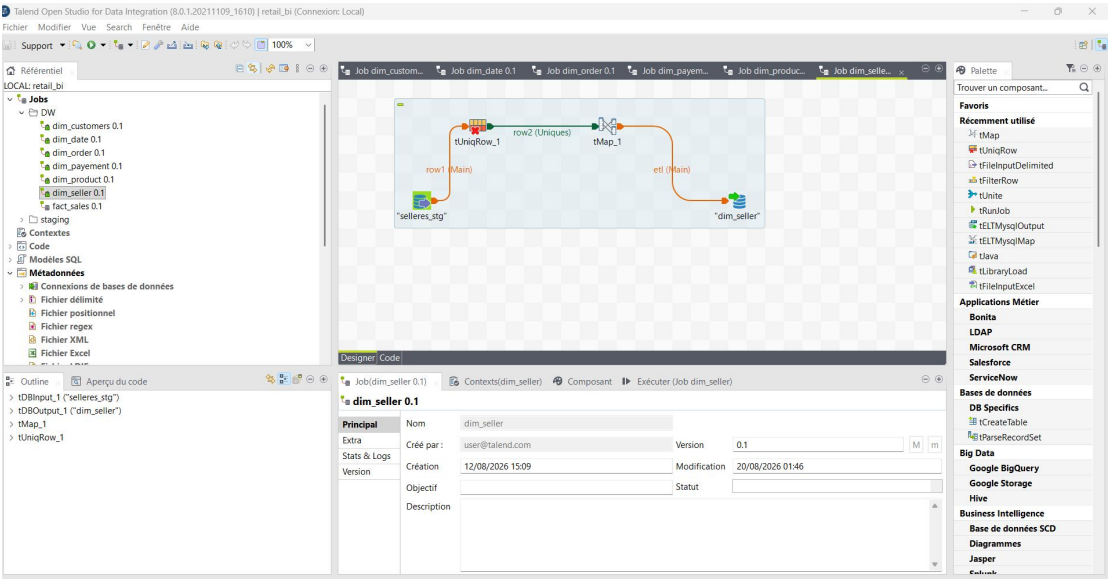
5.Payment dimention etl job



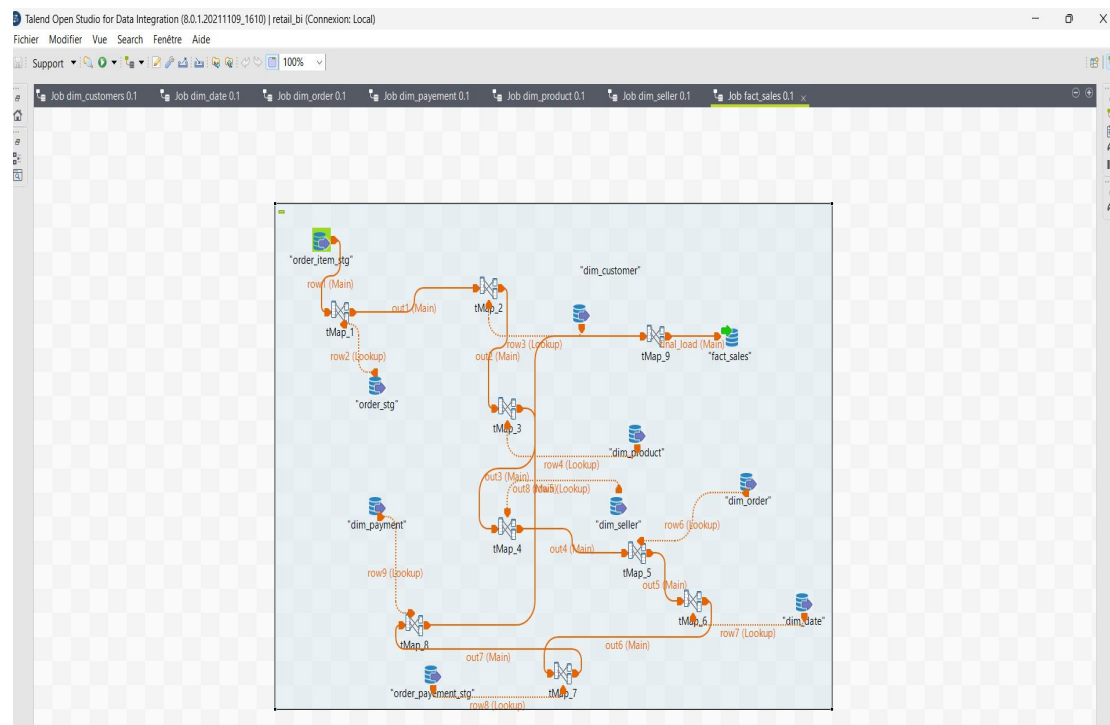
5. Product dimension etl job



6.Seller dimention etl job



7. Fact Sales etl job



Talend Open Studio for Data Integration - tMap - tMap_9

Find:

Var:

Auto map !

out8

Column
payment_key
date_key
order_key
order_id
seller_key
product_key
product_id
customer_key
order_item_id
seller_id
shipping_limit_date
price
freight_value
customer_id
payment_value

final_load

Expression	Column
out8.order_key	order_key
out8.customer_key	customer_key
out8.product_key	product_key
out8.seller_key	seller_key
out8.payment_key	payment_key
out8.date_key.intValue()	date_key
out8.order_item_id	order_item_id
out8.shipping_limit_date	shipping_limit_date
out8.price	price
out8.freight_value	freight_value
out8.payment_value == null ? 0.0f : out8.payment_value	payment_value
out8.price.add(out8.freight_value)	total_item_value

Éditeur de schéma Éditeur d'expression

out8

Colonne	Cle	Type	NL	Modèle de date (C...	Longueur	Précision	Par déf...	Comment...
payment_key		long			19	0	nextval...	
date_key		Integer						
order_key		Long						
order_id		String						
seller_key		Long						
product_key		Integer						

final_load

Colonne	Cle	Type	NL	Modèle de date (C...	Longueur	Précision	Par déf...	Comment...
order_key		long			19	0		
customer_key		long			19	0		
product_key		long			19	0		
seller_key		long			19	0		
payment_key		long			19	0		
date_key		long			19	0		

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Conclusion

The ETL process implemented with Talend successfully transformed the raw Olist datasets into structured and reliable data suitable for Business Intelligence analysis. The developed jobs handled the extraction, transformation, validation, and loading of data into the PostgreSQL environment.

The ETL workflow covered the construction and population of the different dimension tables as well as the `fact_sales` table, ensuring consistency between the staging area and the Data Warehouse. Data type conversions, data cleaning, duplicate handling, and key management were also addressed throughout the process.

